RV INSTITUTE OF TECHNOLOGY AND MANAGEMENT

Sy.No.171/5, Kothanur,8th Phase, JP Nagar, Bengaluru-560076

Software Engineering and Project Management

Course Code: BCS501

2022 Scheme

Assignment on

"Library Management System"

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Semester: CSE V semester

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2025-2026

STEP1: <u>MAKE A LIST OF STAKEHOLDERS FOR THE PROJECT</u>.

Stakeholders of Library Management System

1.Users(Members/Readers):
These are the primary stakeholders who utilize the library's resources and services. They include students, academic professionals, researchers, and general readers who visit the library to borrow, return, or read books through the library management system. Users interact directly with the system for activities such as searching the catalog, reserving books, renewing issued materials, and checking availability. Their satisfaction and ease of access determine the overall effectiveness of the library management system.

2.Librarians/Staff:
Librarians and supporting staff play a crucial role in the day-to-day functioning of the library. They are responsible for managing book issues and returns, updating the catalog, organizing book inventories, and assisting users in navigating the library management system. In the context of an online or digital library management system, technical staff may also be involved in maintaining databases, troubleshooting system issues, and ensuring data integrity. They act as the bridge between the users and the system, ensuring smooth and efficient operations.

3.Administrator:
The administrator is the central authority responsible for overseeing the entire library management system. This role includes monitoring all operations, managing staff accounts, updating system settings, and ensuring that all transactions and data are securely maintained. The administrator ensures proper coordination between users and library staff, enforces policies, and generates analytical reports for decision-making. They hold complete access to system data and functionalities, enabling effective supervision and control of library activities.

STEP2: <u>INTERVIEW EACH STAKEHOLDERS SEPARATELY TO DETERMINE OVERALL WANTS AND NEEDS.</u>

<u>ONLINE - INTERNET ARCHIVE</u>

<u>USERS:</u>

1. How often do you use the online library platform for reading or borrowing books?

Most users engage with the online library on a fairly regular basis, often visiting the platform weekly to explore or borrow digital materials. Some users rely on it almost daily for academic or research-related purposes, while others access it occasionally when they need specific books or references. Overall, the library is seen as a convenient and accessible resource that supports both frequent and occasional readers.

2. **What types of materials do you usually access (e-books, journals, magazines, research papers, etc.)?**

Users typically access a wide range of materials, with e-books and academic journals being the most commonly used. Many also turn to the platform for research papers and study-related content, particularly those involved in higher education or academic projects. A smaller portion of users browse magazines, newspapers, or general reading materials for leisure. The diversity of available resources makes the library useful for both academic and personal purposes.

3. What features do you find most helpful in the online library (e.g., search, book suggestions, download options)?

Users generally appreciate the search and filter tools that make it easier to locate specific materials. The ability to download or borrow books digitally is another feature that many find convenient. Some users value the book recommendation system and reading history, which help them discover new titles or revisit previously accessed content. Overall, features that enhance accessibility and efficiency tend to be the most appreciated.

4. Have you faced any technical issues while using the platform (like login problems, slow loading, or errors)?

While most users find the platform stable, occasional technical issues have been reported. Common problems include slow loading times, login difficulties, and temporary unavailability of certain books or links. These issues are generally minor and infrequent but can sometimes disrupt the reading or borrowing experience. Users suggest that smoother performance and faster page loading would improve overall satisfaction.

5. Do you use the system more for academic, professional, or personal reading purposes?

The online library serves multiple purposes, though academic use remains the most common. Many students and researchers rely on it for accessing study materials, research papers, and scholarly journals. Some users also turn to it for personal reading, exploring fiction, non-fiction, or hobby-related materials. A few professionals use it as a reference tool to support their work. The platform's versatility makes it valuable across different user needs.

6. What do you think about the interface and navigation of the online library (clarity, layout, readability)?

Most users find the interface clean, well-organized, and easy to navigate. The layout supports smooth browsing and reading, with clearly labeled sections and accessible menus. However, a few users feel that the design could be made more modern and visually engaging. Enhancements to layout responsiveness and readability, especially for mobile devices, could make the user experience even better.

7. How do you feel about the speed and performance of the website or app?

Overall, users are satisfied with the platform's performance, though they note that speed and responsiveness can vary. During certain times, the site may experience lag or slow search results, particularly when multiple users are active. Despite these occasional delays, the general experience is smooth, and users appreciate the stability and reliability of the platform.

8. How effective are the filters and search options in helping you find what you need?

The filters and search functions are considered helpful, especially for basic searches by title or author. Users appreciate the ability to sort results by category or relevance. However, many feel that the current filtering options could be more detailed, allowing searches by publication date, subject area, or language. Improving these tools would make locating specific content faster and more precise.

9. What features do you think should be added or improved in the current online library system?

Users believe that introducing advanced search filters, improved mobile compatibility, and better offline reading options would enhance the system's usefulness. Some also suggest adding personalized book recommendations, user reviews, and community features to encourage interaction. A cleaner interface, faster downloads, and enhanced accessibility are common areas identified for improvement.

10. Do you find the digital reading experience (zooming, bookmarking, highlighting) comfortable and efficient?

The digital reading experience is generally viewed as convenient and user-friendly. Users appreciate tools like zooming, bookmarking, and highlighting, which make studying and note-taking easier. However, a few suggest that these functions could be smoother or more responsive, especially on smaller screens. Some users prefer external reading applications for added flexibility, but most find the built-in tools sufficient for everyday reading needs.

IT STAFF

1. How frequently do you monitor or perform maintenance on the online library system?

The IT staff regularly monitor and maintain the online library system to ensure its smooth functioning. Routine checks are typically performed daily to track server health, storage capacity, and network connectivity. Preventive maintenance is scheduled weekly or monthly to apply updates, optimize databases, and verify data integrity. Automated monitoring tools are also used to detect potential performance issues or downtime in real time, allowing quick responses before users are affected.

2. What are the most common technical issues you encounter while maintaining the platform?

Common technical issues include server overloads during peak usage, occasional downtime due to system updates, and database synchronization errors. The team also encounters broken links, outdated metadata, and compatibility challenges when integrating new software components. Sometimes, users report issues related to slow searches or login errors, which are usually traced back to cache or session management problems. While these issues are manageable, they require consistent monitoring and timely troubleshooting to maintain user satisfaction.

3. How do you ensure the system remains secure and protected from unauthorized access or data loss?

Security is maintained through a combination of encryption, access control, and regular audits. The IT staff implement multi-level authentication for administrators, along with strict password and session policies. Data is backed up regularly to secure cloud servers and offline storage systems to prevent loss in case of system failure. Firewalls, intrusion detection tools, and regular vulnerability scans help identify and patch potential security threats. The team also conducts periodic security training and follows best practices for data privacy and compliance.

4. What measures are taken to maintain server uptime and website performance?

To ensure consistent uptime, the team uses load-balancing techniques that distribute network traffic evenly across servers. Regular performance testing helps identify bottlenecks, while caching mechanisms and content delivery networks (CDNs) are used to speed up access for global users. The staff monitor key metrics like CPU usage, memory load, and database response time using performance dashboards. Scheduled maintenance is done during off-peak hours to minimize user disruption, and a backup server setup ensures quick recovery in case of hardware failure.

5. How do you handle large data uploads or backups for the online library?

Large data uploads and backups are handled using automated scripts and batch processing to minimize manual errors. The team schedules these processes during low-traffic periods to reduce server strain. Incremental backups are performed frequently to capture new or updated files without duplicating existing data, while full backups are executed weekly or monthly depending on the data load. Backup files are stored securely in both cloud and physical storage locations, ensuring redundancy and fast recovery in case of system corruption or loss.

6. How user-friendly do you find the admin and backend tools provided for maintenance and monitoring?

The admin and backend tools are generally effective, providing detailed system logs, usage statistics, and access management options. However, some interfaces could be made more intuitive, especially when dealing with complex data queries or configurations. While the tools are powerful, they sometimes require technical expertise to use efficiently. The staff appreciate the automation features for monitoring and reporting, but they also recommend UI improvements and better integration with analytics systems for easier decision-making.

7. What improvements or upgrades do you think are most needed for better performance and reliability?

The IT team believes that upgrading the database management system and optimizing the backend architecture would significantly improve performance. Implementing more scalable cloud infrastructure, enhancing caching mechanisms, and refining the search algorithm could help handle growing user demand. Better automation in maintenance tasks and improved compatibility with new technologies would also reduce manual workload. Additionally, the introduction of AI-based analytics could help in predicting potential system issues before they occur.

8. How do you manage bug reports or technical complaints received from users or librarians?

Bug reports and complaints are tracked through a centralized ticketing or issue-tracking system. Once a report is submitted, it's categorized by severity and assigned to the appropriate technician. Minor issues are often resolved within hours, while more complex problems may require detailed investigation or collaboration with developers. The IT staff maintain communication with librarians and users to update them on progress and resolution timelines. Regular feedback sessions help identify recurring issues and guide future improvements.

9. What challenges do you face in integrating new features or external APIs into the existing system?

Integrating new features or APIs often presents compatibility and stability challenges. The legacy structure of the system sometimes requires code adjustments or version upgrades to support modern tools. There can also be security concerns when connecting third-party services, requiring thorough testing and validation. In addition, data consistency between internal databases and external sources must be carefully maintained. Despite these challenges, the IT staff strive to ensure smooth integration while minimizing downtime and user impact.

10. How do you test the platform after major updates or maintenance tasks to ensure smooth operation?

After any major update or maintenance activity, the IT team performs a structured testing process that includes unit testing, integration testing, and user acceptance testing. Automated testing tools are used to check for broken links, performance issues, or security vulnerabilities. The team also conducts manual testing to verify that core functions like search, login, and downloads are working correctly. Once verified, the updated system is gradually rolled out, and its performance is monitored closely for any anomalies.

ADMINISTRATOR

1. How often do you access the online library's admin panel to oversee operations or user activity?

The administrator accesses the admin panel regularly, to monitor overall system activity, track user engagement, and review content updates. While daily oversight is not always required, frequent checks help ensure that the platform runs smoothly and any issues are identified early. The admin also reviews reports and summaries to stay informed about usage trends and operational health.

2. What are your main responsibilities in managing the online library system?

The administrator's main responsibilities include overseeing the day-to-day functioning of the system, managing user accounts, ensuring content is properly organized, and coordinating with librarians and IT staff. They also approve updates, track system performance, and ensure that the library meets user needs. Essentially, the administrator acts as the central point for decisions related to both the operational and strategic management of the online library

3. How do you monitor system performance, user traffic, and content usage patterns?

Monitoring is done through regular reviews of reports and dashboards that summarize user activity, popular content, and overall system performance. The administrator observes trends such as peak usage times, frequently accessed materials, and any potential bottlenecks in the system. This helps in making informed decisions about resource allocation, content organization, and future improvements to enhance the user experience.

4.What procedures do you follow to handle user account management and permissions?

User account management is handled with careful attention to privacy and access levels. The administrator ensures that users have the appropriate permissions for borrowing, downloading, or accessing restricted content. They also manage librarian and staff accounts, updating roles as needed. Any account issues or requests for access changes are reviewed and resolved promptly to maintain smooth operation and security.

5. How do you coordinate with librarians and IT staff to ensure smooth functioning of the system?

Coordination involves regular communication and planning with both librarians and IT staff. The administrator schedules meetings, shares updates, and ensures that all teams are aligned on system maintenance, content updates, and new feature implementations. Feedback from librarians and IT personnel is incorporated into decision-making to resolve issues efficiently and enhance the library's overall performance.

6. What challenges do you face in maintaining data integrity, user privacy, and compliance with digital policies?

Maintaining data integrity and privacy can be challenging due to the large volume of user accounts and content. The administrator must ensure that user information is secure, records are accurate, and content is properly managed according to copyright and privacy regulations. Keeping up with digital policies and compliance requirements requires continuous attention and occasional updates to processes and procedures.

7. How do you handle server downtimes, software updates, or security incidents?

The administrator works closely with IT staff to manage any interruptions or updates. Planned updates are scheduled during off-peak hours, and contingency plans are in place to reduce disruption. In the event of unexpected downtime or security issues, the administrator coordinates response efforts, communicates with users if necessary, and ensures that services are restored quickly and safely.

8. What features in the current admin dashboard do you find most useful, and which ones need improvement?

The admin dashboard is appreciated for its ability to provide clear overviews of user activity, system alerts, and content statistics. Features like account management and activity reports are particularly helpful. However, some aspects, such as detailed content analytics or predictive insights, could be improved to make decision-making more proactive. A more intuitive layout and better visualization tools would also enhance usability.

9. How do you evaluate user feedback or complaints to guide future updates or feature enhancements?

User feedback and complaints are reviewed regularly to identify patterns or recurring issues. The administrator considers suggestions from both users and librarians to prioritize updates or improvements. Feedback is used to guide decisions on interface enhancements, new features, and content organization, ensuring that the system evolves in a way that meets the needs and expectations of its users.

10. What long-term improvements or expansions would you like to see in the online library system?

Long-term improvements envisioned include better personalization for users, such as recommendations based on reading habits, more interactive features for learning and collaboration, and enhanced mobile accessibility. Expanding the library's content collection and improving search and filtering capabilities are also priorities. Overall, the goal is to make the platform more user-friendly, efficient, and adaptable to the growing needs of both academic and recreational user.

OFFLINE - COLLEGE LIBRARY

USERS

1. How do you usually search for books or study materials in the library?

Most students use the catalog cards or seek assistance from the librarian to locate books by title or subject. Some prefer browsing shelves in their department section as it helps them discover related books they weren't initially searching for.

2. How do you find the process of borrowing and returning books?

They find the process mostly smooth but time-consuming during rush hours since all records are manually logged. Students suggest having a faster issuing process or automated record entry.

3. Are you satisfied with the library's collection of academic books and journals?

Most users appreciate the range of course-related materials, but many mention a shortage of the latest editions of reference books and limited journals for advanced topics.

4. How effective is the library's catalog or index system?

The catalog system is found somewhat outdated since it relies on paper-based indexing. Students suggest implementing a digital catalog accessible through college computers or mobile apps.

5. How do you handle overdue book returns or penalties?

Students mention that manual tracking often leads to confusion about due dates. They prefer receiving reminders through SMS or email notifications.

6. What difficulties do you face while locating books in the library?

Sometimes books are misplaced or wrongly shelved, causing delays. They suggest adding color-coded section labels and shelf maps to make navigation easier.

7. How often do you use the library's reading room?

Most students visit regularly during exams or project work. They find the environment peaceful but would appreciate better ventilation and longer opening hours.

8. Do you think the library should introduce a digital management system?

Nearly all users agree it would save time and reduce human errors. They believe digital systems would improve transparency in book availability and borrowing records.

9. How do you feel about the current library timings and access policies?

Students think the timings should be extended, especially during exams. They also request weekend access for students preparing for competitive exams.

10. What improvements do you suggest for better user experience?

They recommend installing computers for online catalog searches, introducing an automated issuing system, and setting up a feedback kiosk to share user opinions directly.

<u>LIBRARIAN</u>

1. How do you currently manage book inventories and records?

The librarian maintains registers for book entries, issues, and returns manually. It is efficient for small volumes, but tracking and verifying thousands of entries is challenging without digital assistance.

2. What are the common challenges faced in managing the library's collection?

Misplaced books, manual errors during entry, and difficulty in updating availability status are frequent issues. Handling late return records manually also consumes significant time.

3. How do you assist students in finding the required materials?

The librarian uses subject-based classification and personal experience to guide students. However, searching becomes slow when students request specific editions or rare books.

4. What methods do you use to track issued and returned books?

Entries are maintained in registers, but this makes it difficult to generate reports on popular books or borrowing patterns. A computerized system could automate these reports easily.

5. How do you manage overdue books and fines?

Currently, fines are calculated manually based on due dates. Errors sometimes occur, leading to disputes. A digital system could automatically calculate and record fines.

6. What role do you play in maintaining the library catalog?

The librarian updates the card catalog regularly, ensuring correct author names and book placements. Still, manual updates are time-consuming compared to automatic database entries.

7. How do you evaluate the demand for new books or journals?

Requests are gathered from students and faculty through suggestion slips. Decisions are made based on frequency of requests and curriculum relevance, but data analysis is manual.

8. How do you ensure proper book arrangement and classification?

Books are arranged based on Dewey Decimal Classification, but students sometimes misplace them. A barcode-based tracking system could improve monitoring.

9. What improvements do you think would make library management more efficient?

Introducing library management software with barcode scanning, digital catalog access, and online book requests would reduce workload and errors significantly.

10. How do you envision the library in the next five years?

The librarian imagines a hybrid system with both physical and digital access, enabling users to search, reserve, and renew books online while maintaining the charm of the offline reading environment.

ADMINISTRATOR

1. How do you oversee the overall management of the library?

The administrator monitors resource allocation, book procurement budgets, and staff coordination. Reports are collected manually from the librarian, making data analysis time-consuming.

2. What challenges do you face in maintaining library records?

Manual data storage results in incomplete or delayed information on book circulation and purchases. Auditing old records is especially difficult without a digital backup.

3. How do you handle budgeting for library resources?

The administrator allocates funds based on past records and departmental requirements. However, manual tracking makes it difficult to predict which books are most in demand.

4.How are procurement and inventory updates currently handled?

Book orders and updates are done through physical forms and vendor communication. Implementing a digital procurement log would make the process more transparent and traceable.

5. How do you ensure efficient communication between students, librarians, and management?

Communication mainly happens through in-person meetings or notice boards. An integrated library management system could streamline communication via notifications or digital requests.

6. How do you monitor the performance of library services?

Feedback is collected manually through suggestion boxes and verbal discussions. A digital feedback module could provide more structured insights.

7. What policies exist for maintaining or upgrading library systems?

Policies emphasize manual auditing and periodic review of inventory. The administrator acknowledges the need for digital policy frameworks to ensure accountability and accuracy.

8. How do you handle data related to book circulation and usage patterns?

Currently, data compilation takes days as it depends on paper records. Automated analytics would make it easier to identify usage trends and plan resource allocation.

9. What are your plans for modernizing the college library?

Plans include introducing an integrated digital library system with barcode scanners, online catalogs, and digital membership cards for students and staff.

10. How do you envision the library's role in enhancing student learning?

The administrator sees the library as a vital academic hub. By digitizing operations, students would get faster access to materials, and the library could become a knowledge center that complements classroom learning.

HYBRID - JUST BOOKS

USERS

1. How easy is it to search for and locate books using the current library system?

Most users find the search system moderately helpful but outdated. A few mention that while books are listed by category, the search interface isn't always intuitive—especially when they look for books by partial titles or author names. Others rely on librarians for assistance because the database sometimes misses recently added titles. Users appreciate when books are displayed on shelves corresponding to system categories but wish the digital search could be faster and more responsive.

2. How do you usually check the availability of a book before visiting the library?

Several users say they use the online catalogue link provided by the library, but it often doesn't reflect real-time availability. Many still prefer calling the library directly. A few regular readers mention maintaining a wishlist in the library’s app, but others didn’t even know this feature existed, showing uneven awareness and usage of the system.

3. How satisfied are you with the book issue and return process in the system?

Users describe the issuing process as smooth overall but slightly time-consuming during rush hours. Some appreciate the barcode scanning system, while others face delays due to manual entry when the scanner malfunctions. They like receiving due-date reminders via SMS but feel penalty updates could be more transparent in the portal.

4. Have you experienced any difficulties with late fees or renewal options in the system?

Many users report confusion when late fees don’t immediately update in their profiles. Renewals through the system often fail or require librarian assistance. Some wish for an automatic renewal feature or notifications before the due date, saying that missed reminders are a common problem.

5. Do you find the recommendation or suggestion features useful?

Responses are mixed—some readers love personalized recommendations that align with their reading history, while others think the suggestions are too generic or repetitive. A few mention wanting more localized suggestions based on Indian authors and genres popular in their region.

6. How often do you face system downtime or errors while using the catalog?

Most users agree the system works fine on normal days but slows down during weekends when more people log in. Some mention occasional server errors that prevent them from checking availability, while a few say they rarely face any issues because they mostly visit in person.

7. Do you feel your borrowing history and reading data are used meaningfully?

A few users appreciate getting stats about their reading habits, like “most borrowed genres,” but most don’t see much benefit from stored data. Some express privacy concerns, unsure if their data is shared for marketing. Others would like the data to be used to improve reading suggestions or event invites.

8. How helpful do you find the notifications and reminders in the system?

Users like SMS reminders for due dates but complain that renewal notifications sometimes arrive late or not at all. Some mention that push notifications on the app don’t work on all phones. Overall, they appreciate the reminders but request consistency and better timing.

9. How easy is it to pay fines or renew memberships through the system?

Several users still pay fines manually at the counter. The online payment gateway works for some but not all devices, leading to frustration. A few say UPI options would make the process smoother. Membership renewals often require a physical visit, which users find inconvenient.

10. What improvements would you suggest for the system?

Most users recommend a faster search engine, real-time updates on availability, online fine payments, and improved notifications. A few also suggest integrating the app with e-books or offering delivery tracking for home deliveries under the premium plan.

LIBRARIANS

1. How easy is it to search for and locate books using the current library system?

Librarians say the system helps them manage daily operations but has several usability issues. Advanced searches sometimes return too many irrelevant results. They note that updating new arrivals or removing lost books from the database takes extra time, especially when internet connectivity is weak.

2. How do you usually check the availability of a book before users request it?

Librarians rely on the system's backend interface, which displays stock and borrower details. However, they mention discrepancies between the digital record and the physical shelves due to delayed updates. Some keep manual checklists to ensure accuracy.

3. How satisfied are you with the book issue and return process in the system?

They find barcode scanning reliable, though the system occasionally freezes. Bulk returns after weekends are challenging as each transaction must be processed individually. They appreciate the logging accuracy but wish for faster batch processing.

4. Have you experienced any difficulties with late fees or renewal options in the system?

Librarians report that automatic fine calculations sometimes glitch, showing incorrect amounts. Renewal requests often don't sync instantly, forcing them to adjust entries manually. They prefer more automation to reduce manual corrections.

5. Do you find the recommendation or suggestion features useful?

They believe the recommendation module could help engage users better but say the system doesn't utilize user history effectively. Some librarians manually curate book displays based on borrowing patterns since the automated suggestions are limited.

6. How often do you face system downtime or errors while managing inventory?

All librarians mention occasional downtimes, particularly during inventory updates or when too many terminals are active. They keep offline logs as backups to avoid losing track of borrowed books.

7. Do you feel user data and transaction logs are being managed effectively?

They trust the database but feel more analytics could be added—for example, reports on trending books or inactive users. Data export features exist but often misformat records, creating extra work.

8. How helpful do you find the notifications and reminders in the system?

Librarians value the automatic reminder feature because it reduces follow-up calls, but they also notice inconsistencies—some users complain about missed alerts. They occasionally send manual reminders to ensure returns.

9. How easy is it to process fines or membership renewals through the system?

Processing fines is fairly simple, but payments often have to be manually verified. Membership renewals are tracked digitally, though the system doesn't send reminders consistently, requiring staff to follow up in person.

10. What improvements would you suggest for the system?

Librarians unanimously request a smoother backend, faster updates, better data synchronization, and integration with online payment platforms. They also suggest advanced analytics to predict book demand and improve stock management.

ADMINISTRATOR

1. How easy is it to search for and locate books using the current library system?

The administrator views the system as functional but outdated. They note that the catalog lacks AI-based filtering or predictive search. Plans to implement a more dynamic interface are underway, but budget constraints delay upgrades.

2. How do you usually verify the availability or stock updates of books?

The administrator monitors availability through daily system-generated reports and librarian feedback. They acknowledge occasional mismatches between system data and shelf reality, attributing them to delayed manual updates.

3. How satisfied are you with the issue and return management process?

They believe the current process is efficient for an offline setup but wish to automate repetitive librarian tasks. Integrating RFID tags or a faster barcode module is on their roadmap to minimize queue times.

4. Have you experienced difficulties with fine or renewal management?

From a managerial standpoint, the system's fine module needs refinement—especially in syncing transactions. The admin often audits logs to ensure financial accuracy and transparency.

5. Do you find the recommendation or suggestion algorithm effective?

The administrator feels it’s underutilized. They want to enhance it using borrowing trends and member demographics to encourage more engagement and book circulation.

6. How often do you face technical issues or downtime?

They mention system downtime mainly occurs during maintenance or server upgrades. Although backups exist, they want to move towards a hybrid (cloud + local) infrastructure for stability.

7. Do you think data is being managed and used effectively?

They emphasize the need for advanced analytics dashboards to track trends, late-return frequency, and member engagement. While the data is secure, it’s not yet being leveraged for strategic decisions.

8. How helpful do you find the notifications and reminders system?

The administrator recognizes the benefit but notes inconsistencies across communication channels. They are considering integrating WhatsApp notifications for higher engagement.

9. How efficient are fine and membership payment modules from your perspective?

They see them as semi-automated but functional. They aim to implement UPI and auto-renewal systems to make payments easier and reduce staff workload.

10. What improvements would you suggest for the library management system?

The administrator envisions a fully integrated, mobile-compatible system with real-time inventory, automated renewals, better analytics, and smoother user experience. They also emphasize cybersecurity and scalability for future branches.

STEP3: <u>BUILD A PRELIMINARY LIST OF FUNCTIONS AND FEATURES BASED ON STAKEHOLDERS INPUT</u>

1. User Access & Account Management

Key Features:

- Secure login and authentication
- Profile dashboard (borrowed books, due dates, reading history)
- Digital membership card
- Multi-role access (Student, Librarian, Administrator)

Requested By:
Users (Online & Offline), Administrators

2. Book Search & Discovery

Key Features:

- Advanced search (by title, author, publication year, category, language)
- Smart filters and sorting options
- Auto-suggest search results
- Personalized book recommendations based on reading history
- Unified catalog: search across both physical and digital inventories

Requested By:
Users (Online + Hybrid), Librarians, IT Staff

3. Catalog & Book Information

Key Features:

- Detailed book information (availability, edition, shelf/location number)
- Barcode/QR code scanning for physical books
- Real-time stock and location updates
- Display of cover images, user reviews, and ratings

Requested By:
Users, Librarians

4. Borrowing & Reservation System

Key Features:

- Digital issue and return record management
- Book reservation or hold request functionality
- Renewal requests (online and offline synchronization)
- Automatic due-date tracking
- Overdue fine calculation and notification system

Requested By:
Users, Librarians, Administrators

5. Digital Reading Features

Key Features:

- Integrated e-book viewer with:
  - Zoom
  - Highlighting and note-taking

    - Bookmarking
    - Download or restricted offline mode
- Mobile and tablet compatibility

Requested By:
Online Users

6. Performance & Reliability Enhancements

Key Features:

- Load balancing and optimization during peak usage hours
- Fast search performance using indexing and caching
- Cloud-based distributed storage system
- Real-time error handling and downtime alerts

Requested By:
IT Staff (Primary)

7. Content Management (Librarian Tools)

Key Features:

- Digital catalog entry and updates
- Barcode generation for new books
- Bulk upload of book records
- Book misplacement tracker (last known scan location)
- Inventory reports and audit logs

Requested By:
Librarians, Administrators

8. Security & Data Protection

Key Features:

- End-to-end encryption for user and content data
- Multi-level access control for different user roles
- Regular data backups (incremental and full)
- Real-time monitoring and intrusion detection

Requested By:
IT Staff, Administrators

9. Analytics & Usage Insights

Key Features:

- Analysis of popular books and peak usage hours
- User activity and engagement reports
- Predictive analytics for book procurement and system upgrades

Requested By:
Administrators, Librarians

10. Communication & Feedback System

Key Features:

- In-app notifications (due dates, holds, reminders)
- Email and SMS alerts for users
- Integrated feedback form or kiosk interface
- Automated responses for frequently asked questions

Requested By:
Users, Administrators

11. Accessibility & UI Enhancements

Key Features:

- Modern and responsive interface
- Mobile app and adaptive layout design
- Clear section labeling with shelf maps
- Visually assistive options (font size control, high-contrast mode)

Requested By:
Users (Online & Offline)

STEP4:SCHEDULE A SERIES OF FACILITATED APPLICATION SPECIFICATION

Library Type	Stakeholders Involved	Date	Purpose of Meeting
Online Library System	Users (Students), IT Staff, Administrators	10 September 2025	Collect user expectations regarding online search, login system, e-books, security, and system performance.
Offline Library System	Users (Students), Librarian, Administrators	11 September 2025	Identify issues in book issue/return, manual cataloging, shelf location, and fine management
Hybrid Library System	Hybrid Users, Administrators	12 September 2025	Understand integration needs between online discovery and offline book borrowing, including real-time availability.

STEP 5:CONDUCT MEETINGS

1. Online Library – Meetings Conducted

- A facilitated application specification meeting for the Online Library System was conducted on 10 September 2025 with three stakeholder groups: Users, IT staff, and Administrators.
- Users expressed that they prefer the online library because it allows easy book searching, login-based access, and time-saving compared to offline libraries.
- They also requested additional features such as e-books, browsing for newly published books, and extending borrowing time.
- The IT staff discussed concerns such as system maintenance, monitoring the online library regularly, resolving errors, and ensuring no unauthorized access to book content.
- Administrators highlighted the need for secure user access control, restriction on downloading e-books, timely password updates, and theft-prevention measures.
- All points were documented for requirement finalization.

2. Offline Library – Meetings Conducted

- The requirement gathering meeting for the Offline Library System was conducted on 11 September 2025 with stakeholder participation from Offline Users, Librarians, and Administrators.
- Offline library users mentioned delays during the book issue/return process during peak hours and difficulty in checking book availability.
- They requested a system showing exact shelf location and immediate status updates for misplaced books.
- Librarians emphasized the need for a centralized catalog to track all book details such as number of books, authors, location, and type of content, along with automated issuing and returning.
- Administrators requested access controls, proper role management, and a systematic record of all borrowed and returned books.
- These insights highlighted major areas requiring digital improvement in the offline library workflow.

3. Hybrid Library – Meetings Conducted

- A separate hybrid library meeting was arranged on 12 September 2025 involving Hybrid Users and Administrators.
- Users shared that hybrid libraries give the advantage of both online and offline access, enabling them to search for books online and later collect them from the library.
- They suggested adding features such as a quick search engine, simple catalog navigation, user-friendly interface, and recommendations of newly published books of interest.
- Administrators requested restrictions on unauthorized copying, automatic fine calculation, and managing records for both physical and digital books together.
- The input received confirmed the requirement for a unified Library Management System supporting both digital access and physical book handling.

STEP 6: PRODUCE INFORMAL USER SCENARIOS AS PART OF EACH MEETING

Informal User Scenarios – Online Library System

1. Student Login & Book Search
A student logs in using their username and password from home. They search for a newly published programming book and check if an e-book version is available to read immediately without going to the library.

2. IT Staff Monitoring System Errors
A system maintenance technician receives an alert about slow page loading during peak class hours. They quickly check the online library system to resolve the issue before more students face delays.

3. Administrator Restricting Unauthorized Access
The admin notices multiple failed login attempts from unknown users. They immediately apply password reset policies and restrict suspicious access to prevent e-book theft.

<u>Informal User Scenarios – Offline Library System</u>

1. User Searching & Issuing a Book
A student walks into the library during lunch break and stands in a long queue to return and borrow books. They struggle finding a book because the shelf location is unclear, causing delay for others behind.

2. Librarian Managing Records Manually
The librarian updates the physical register and checks the due date manually for each book issued. If a book is misplaced, they spend extra time searching shelves due to lack of tracking.

3. Admin Handling Fine Calculation
An administrator reviews overdue cases using handwritten notes. They manually calculate the fine amount and update the student record, which sometimes leads to errors.

<u>Informal User Scenarios – Hybrid Library System</u>

1. User Reserving Online and Collecting Offline
A regular user searches a novel online, finds it available, reserves it instantly, and visits the library to pick it up later without waiting in a queue.

2. Admin Managing Both Physical & Digital Records
The admin checks that an e-book borrowed online and a physical magazine borrowed offline are both updated in the same system to avoid duplication or missing details.

3. User Discovering New Books of Interest
A user browsing online receives recommendations for newly published comics. They check if the physical copies are available and decide to visit the library to borrow them.

STEP 7: REFINE USER SCENARIOS BASED ON STAKEHOLDER FEEDBACK

1. OFFLINE LIBRARY

Informal Scenario(Pain Point)	Refined Scenario Focus(Goal)	Implied Requirements Category
User Searching and Issuing a Book (Finding books takes too long ,creating queues.)	**Goal:** Enable users to efficiently locate and reserve physical books before approaching the circulation desk**.**	**Location Management & Hold System** (Functional)
Librarian Managing Records Manually (Frequent errors and no automated follow-up.)	**Goal:** Automate all circulation tasks, from check-in/out to overdue notices and fine calculation.	**Automated Circulation Module** (Functional)
Admin Handling Fine Calculation(Manual fine calculation leads to errors and poor auditing.)	**Goal:** Ensure all fines are calculated automatically, immediately applied to user accounts, and logged for administrative auditing.	**Financial & Audit Logging** (Functional / Non-Functional)

2. ONLINE LIBRARY

Informal Scenario (Pain Point)	Refined Scenario Focus (Goal)	Implied Requirements Category
Student Login & Book Search (Slow page loading during peak hours.)	**Goal:** Guarantee consistent search and access speed, even when a large number of students are accessing e-resources concurrently.	**System Performance & Scalability** (Non-Functional)
IT Staff Monitoring System Errors (Need to proactively address issues before user complaints.)	**Goal:** Implement real-time monitoring and alerting for performance bottlenecks (e.g., slow database queries or network latency).	**Maintainability & Monitoring** (Non-Functional)
Administrator Restricting Unauthorized Access (Need to prevent e-book theft and account compromise.)	**Goal:** Enforce strict, automated security policies, including **multi-factor authentication (MFA)** and **IP-based access restrictions**.	**Security & Authentication** (Non-Functional)

3. HYBRID LIBRARY

Informal Scenario (Pain Point)	Refined Scenario Focus (Goal)	Implied Requirements Category
User Reserving Online and Collecting Offline (Need for a single process for different asset types.)	**Goal:** Create a **Unified Reservation/Hold Module** that can handle both physical pickup reservations and immediate digital access grants.	**Unified Interface Logic** (Functional)
Admin Managing Both Physical & Digital Records (Risk of data silos and reporting inconsistencies.)	**Goal:** Ensure all physical and digital records reside within a **single, logically unified database schema** to enable comprehensive reporting.	**Data Integrity & Consistency** (Non-Functional)
User Discovering New Books of Interest (Recommendations should consider all available formats.)	**Goal:** Implement a **recommendation engine** that cross-references user preference with both physical inventory and digital licenses.	**Usability & Recommendation Engine** (Functional)

STEP 8: BUILD A REVISED LIST A STAKEHOLDER REQUIREMENTS

1. Online Library Requirements

A. Functional Requirements

- **Advanced Book Search:** Users must be able to search by **title, author, category, and publication year**.
- **Real-time Availability:** The system must display the **accurate real-time availability** of e-books and journals.
- **User Account Management:** Includes features for **login, viewing borrow history, managing reservations, and profile updates**.
- **Automated Digital Access:** The system must provide **automated download and access** for digital content.
- **Recommendation System:** Must suggest **new or relevant e-books** based on user preferences.
- **Notifications:** Must send alerts for **due dates, overdue materials, and new arrivals**.
- **Admin Features:** Tools to **manage content, monitor system, and generate usage reports**.

B. Non-Functional Requirements

- **Performance/Scalability:** Must ensure **consistent performance** even with high user load.

- **Security:** Requires **secure authentication** (multi-factor) and **encryption** for e-content.
- **Reliability:** The system must have **minimal downtime**.
- **Data Integrity:** Requires high **data integrity** for all digital assets.
- **Usability:** Needs an **intuitive and responsive UI** for web and mobile users.
- **Maintainability:** Must include **real-time monitoring tools** for IT staff.

2. Offline Library Requirements

A. Functional Requirements

- **Efficient Search & Retrieval:** Tools for efficient search and retrieval of physical books (via catalog cards or a digital catalog).
- **Automated Circulation:** The system must handle **automated check-in, check-out, overdue tracking, and fine calculation**.
- **User Account Integration:** User accounts must be integrated to track **borrowing history and reservations**.
- **Book Tracking:** Must use **Barcode or QR-based tracking** of book location and movement.
- **Notifications:** Must send alerts for **due dates and fines via SMS/email**.
- **Librarian Tools:** Tools for **inventory updates** and **misplaced book tracking**.
- **Admin Features:** Includes **audit logs, resource allocation, and collection updates**.

B. Non-Functional Requirements

- **Performance:** The system must have **fast response** even during peak hours.
- **Security:** Requires **restricted access** to admin and librarian accounts.
- **Reliability/Backup:** Needs a **reliable backup system** for physical inventory data.
- **Data Consistency:** Requires **data consistency** between catalog records and actual book locations.
- **Usability:** Needs a **usable interface** for librarians and staff.
- **Maintainability:** Must include **maintainability and error logging** for IT staff.

3. Hybrid Library

A. Functional Requirements

- **Unified Search System:** A single system for searching **both physical and digital books**.
- **Integrated Reservation System:** A reservation/hold system that integrates **online reservations with offline collection**.
- **Automated Issue/Return:** Automated issue/return for physical books that immediately trigger **online updates**.
- **Recommendation System:** Must recommend materials considering **both physical and digital availability**.
- **Notifications:** Notifications for **reservations, due dates, and availability updates**.
- **Admin Tools:** Provides **unified reporting, audit logs, and catalog synchronization**.

- **Librarian Tools:** Tools to manage **physical book placement** and update the online catalog.

B. Non-Functional Requirements

- **Performance/Scalability:** Must be **scalable** to handle both online and offline users seamlessly.
- **Security:** Requires comprehensive **security** for both digital and physical inventory data.
- **Reliability:** Needs high system **reliability and uptime** for hybrid operations.
- **Data Integrity:** Requires strict **data integrity** between physical and digital systems.
- **Usability:** Needs a **user-friendly interface** for students, librarians, and administrators.
- **Maintainability:** Must include **monitoring and maintainability** for IT staff across all hybrid components.

STEP 9: USE QUALITY FUNCTION DEPLOYEMENT TECHNIQUES TO PRIORITIZE REQUIREMENTS

Step 1: Identify Requirements (Normal / Expected / Exciting)

- Normal Requirements: Explicitly requested by stakeholders; must be present.
- Expected Requirements: Implicit, fundamental; absence causes dissatisfaction.
- Exciting Requirements: Go beyond expectations; delight the users.

1. Online Library

Type	Requirement	Description
Normal	Login / Logout	Users can securely access the platform.
Normal	Search & Borrow e-books	Locate, borrow, and download e-books efficiently.
Expected	System uptime & reliability	Smooth operation during peak hours; minimal downtime.
Expected	Security	Prevent unauthorized access; MFA for sensitive actions.
Exciting	Personalized recommendations	AI-based suggestions based on reading history.
Exciting	Notifications & alerts	Smart notifications for due dates, new arrivals, or events.

2. Offline Library

Type	Requirement	Description
Normal	Unified catalog	Single interface for physical and digital books.
Normal	Online reservation	Reserve physical books online for pickup.
Expected	Payment/fine integration	Smooth fine management and payment process.
Expected	Notifications	Alerts for reservations, due dates, and fines.
Exciting	AI recommendations	Personalized suggestions across digital + physical inventory.
Exciting	Admin hybrid dashboard	Centralized control of both offline and online resources.

3. Hybrid Library

Type	**Requirement**	**Description**
Normal	Catalog search	Quickly locate physical books via catalog or system.
Normal	Automated check-in/out	Reduce manual errors and speed up book circulation.
Expected	Fine calculation & overdue alerts	Automatically calculate fines and notify users.
Expected	Data integrity & backups	Ensure accurate, secure records.
Exciting	Book usage analytics	Insights on popular books and borrowing trends.
Exciting	Extend library hours access	Flexible access for students during exams or projects.

Step 2: Assign Importance and Correlation (Prioritize)

- Importance (1–5): How critical the requirement is to users.
- Correlation (1–5): Strength of relationship between requirement and satisfaction.
- Priority Score = Importance × Correlation

1.Online Library Prioritization

Requirement	Type	Importance	Correlation	Priority Score
Login / Logout	Normal	5	5	25
Search & Borrow	Normal	5	5	25
System uptime	Expected	5	5	25
Security	Expected	5	5	25
Personalized recommendations	Exciting	4	4	16
Notifications	Exciting	4	5	20

2.Offline Library Prioritization

Requirement	Type	Importance	Correlation	Priority Score
Catalog search	Normal	5	5	25
Automated check-in/out	Normal	5	5	25
Fine calculation & alerts	Expected	5	5	25
Data integrity & backups	Expected	5	5	25
Book usage analytics	Exciting	4	4	16
Extended library hours	Exciting	3	3	9

3.Hybrid Library Prioritization

Requirement	Type	Importance	Correlation	Priority Score
Unified catalog	Normal	5	5	25
Online reservation	Normal	5	5	25
Payment/fine integration	Expected	5	5	25
Notifications	Expected	4	5	20
AI recommendations	Exciting	4	4	16
Admin hybrid dashboard	Exciting	5	5	25

STEP 10: PACKAGE REQUIREMENTS FOR INCREMENTAL DELIVERY

1.Online Library

- Increment 1: Login/Logout, Search & Borrow, System uptime
- Increment 2: Security, Notifications
- Increment 3: Personalized Recommendations

2.Offline Library

- Increment 1: Catalog search, Automated check-in/out, Fine calculation & alerts
- Increment 2: Data integrity & backups
- Increment 3: Book usage analytics, Extended library hours

3.Hybrid Library

- Increment 1: Unified catalog, Online reservation.
- Increment 2: Payment/fine integration, Notification.
- Increment 3: AI recommendations, Admin hybrid dashboard

STEP 11: NOTE CONSTRAINTS AND RESTRICTIONS THAT WILL BE PLACED ON THE SYSTEM

1. Online Library Constraints

Constraint / Restriction	Description / Reason
Platform compatibility	Must work on web browsers and mobile devices.
Security compliance	Must follow authentication standards (MFA, password policies).
Bandwidth limitations	Performance should remain acceptable for low-speed internet users.
Data privacy	Personal user information must comply with IT/data protection policies.
Digital license restrictions	Access to e-books must honor licensing agreements and DRM.
Maximum concurrent users	System should handle peak loads without crashing.

2. Offline Library Constraints

Constraint / Restriction	Description / Reason
Hardware dependency	System must work with existing library computers and barcode scanners.
Space limitation	Physical books must fit within available shelving; automated check-in/out must account for limited desk space.
Backup frequency	Must maintain regular backups to prevent data loss due to hardware failure.
Manual intervention	Some processes (like shelving or physical book handling) cannot be automated.
Operational hours	System availability limited to library opening hours.
Fine enforcement	Fines should follow legal or institutional rules and cannot be overridden arbitrarily.

3. Hybrid Library Constraints

Constraint / Restriction	Description / Reason
Integration compatibility	Online and offline modules must share a single database schema.
Real-time updates	Reservations or checkouts must synchronize immediately between online and offline modules.
Security & access control	Users should not bypass system rules to reserve, borrow, or pay fines.
Data integrity	Prevent conflicts between digital and physical records.
Licensing & copyright	Digital and physical content access must follow legal licensing agreements.
System downtime	Planned maintenance must minimize disruption for both online and offline users.

STEP 12: DISCUSS METHODS FOR VALIDATING THE SYSTEM

This plan outlines how the system can be validated in the future to ensure it meets stakeholder requirements, performs reliably, and adheres to constraints.

1. Online Library Validation Methods

Validation Method	Description / Purpose
User Acceptance Testing (UAT)	Involve students, researchers, and faculty to test searching, borrowing, and e-book access. Verify if the system meets usability and accessibility requirements.
Functional Testing	Test search, filter, download, recommendation engine, and account management to ensure all features work as specified.
Performance Testing	Check system response time, load handling, and uptime for multiple simultaneous users.
Security Testing	Verify authentication, authorization, data privacy, and protection against hacking attempts.
Cross-platform Testing	Ensure compatibility across browsers, mobile devices, and OS versions.

2. Offline Library Validation Methods

Validation Method	**Description / Purpose**
Manual Scenario Testing	Simulate book issuing, returns, fines, and shelf management to check correctness.
Functional Testing	Validate circulation module, catalog updates, and barcode scanning.
Audit Testing	Check record accuracy, fine calculations, and reporting.
Load Testing	Verify performance during peak hours (exam periods or library rush).
Backup & Recovery Testing	Ensure proper backup of book records and recovery after system failures.

3. Hybrid Library Validation Methods

Validation Method	Description / Purpose
End-to-End Testing	Validate seamless interaction between online and offline modules (e.g., online reservation → offline pickup).
Integration Testing	Verify data consistency between digital and physical inventory.
User Acceptance Testing (UAT)	Gather feedback from both online and offline users to ensure unified experience.
Security & Compliance Testing	Ensure licensing restrictions, access control, and copyright rules are enforced.
Performance & Synchronization Testing	Test real-time updates, concurrent access, and responsiveness across all modules.